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A simple discovery is shaking the foundations of spintronics

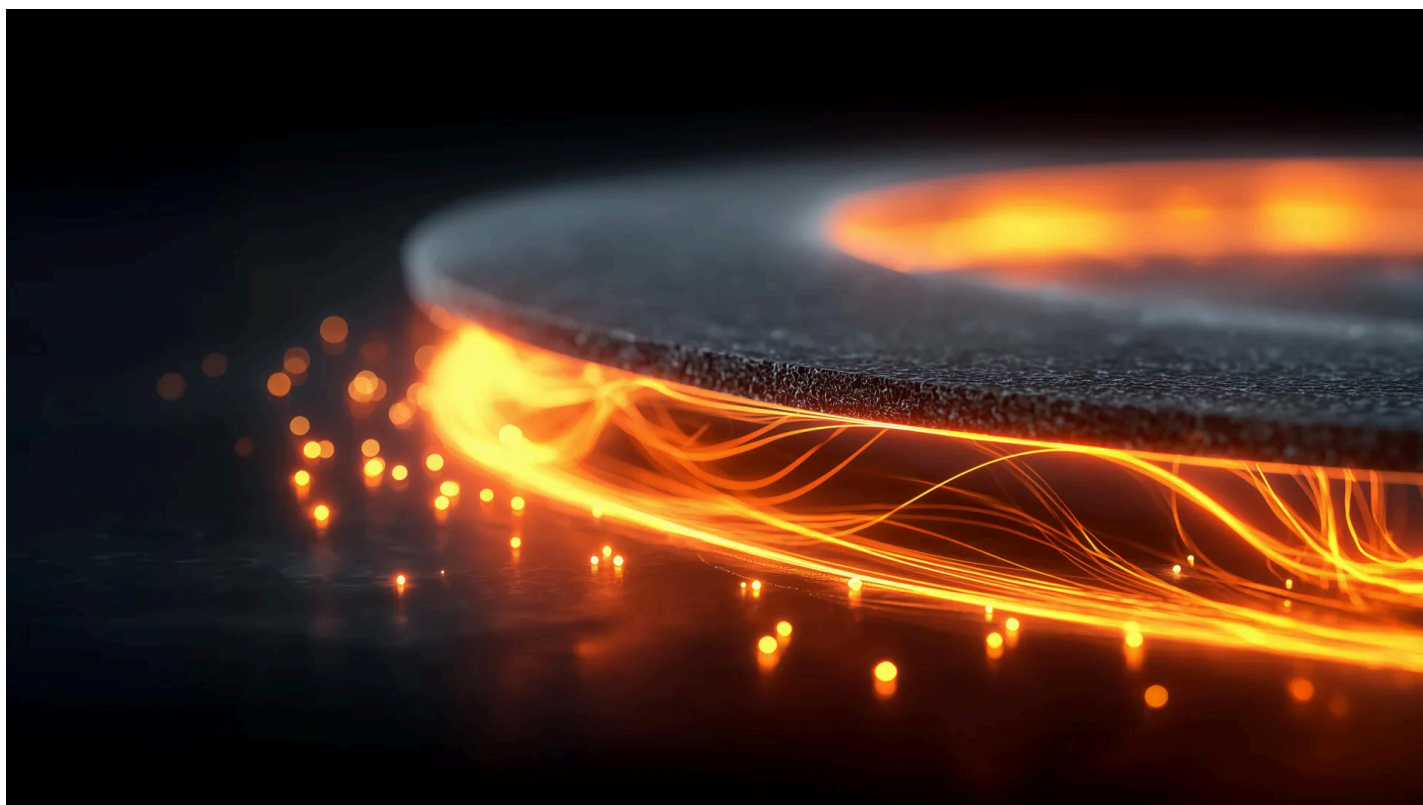
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Summary: A long-standing mystery in spintronics has just been shaken up. A strange electrical effect called unusual magnetoresistance shows up almost everywhere scientists look—even in systems where the leading explanation, spin Hall magnetoresistance, shouldn't work at all. Now, new experiments reveal a far simpler origin: the way electrons scatter at material interfaces under the combined influence of magnetization and an electric field.

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FULL STORY



A baffling spintronics effect just got a simpler explanation—and it upends one of the field's most trusted theories. Credit: AI/ScienceDaily.com

One of the most striking findings in spintronics is unusual magnetoresistance (UMR). In this effect, the electrical resistance of a heavy metal changes when it is placed next to a magnetic insulator and the direction of magnetization rotates within a plane perpendicular to the flow of electric current. This behavior played a key role in shaping the concept of spin Hall magnetoresistance (SMR), which became the dominant explanation for UMR. Over time, SMR has been widely used to interpret results from many types of experiments, including magnetoresistance measurements, spin-torque ferromagnetic resonance, harmonic Hall voltage studies, magnetic field sensors, and switching of magnetization or Néel vectors.

As more experiments were carried out, researchers noticed something puzzling. UMR showed up in nearly all magnetic systems, even in cases where no spin Hall material was present. The effect was also detected in systems where SMR theory clearly does not apply (e.g., those without a spin Hall effect). To address these inconsistencies, scientists proposed a growing list of alternative explanations tied to spin currents or related effects. These included Rashba-Edelstein MR, spin-orbit MR, anomalous Hall MR, orbital Hall MR, crystal-symmetry MR, orbital Rashba-Edelstein MR, and Hanle MR. Each was designed to account for the "SMR-like" signals observed in specific experimental setups.

A New Experimental Answer Emerges

More recently, Prof. Lijun Zhu of the Institute of Semiconductors at the Chinese Academy of Sciences and Prof. Xiangrong Wang of the Chinese University of Hong Kong presented clear experimental evidence pointing to a different origin of universal UMR. Their work shows that the effect arises from how electrons scatter at interfaces, with this scattering controlled by both the magnetization and the electric field at the interface. This process is known as two-vector magnetoresistance. Crucially, this explanation does not rely on spin currents, which removes many of the complications found in earlier models.

Their experiments revealed that very large UMR signals can appear even in single-layer magnetic metals. They also found that the effect includes higher-order contributions and follows a universal sum rule. All of these observations closely match what the two-vector MR model predicts, without the need to invoke spin-current-based mechanisms.

Reinterpreting Decades of Experimental Data

The researchers also carried out a careful review of previous studies. This reanalysis showed that many influential experimental results once attributed to spin Hall magnetoresistance or other spin-current-related, or even unrelated, mechanisms can be consistently explained using the two-vector MR framework. In addition, they highlighted several experimental and theoretical findings that directly conflict with spin-current-based MR models but are naturally explained by the two-vector approach.

A Challenge to a Long-Standing Theory

Together, these results pose a serious challenge to the long-accepted SMR theory. They provide the first strong experimental confirmation of the two-vector magnetoresistance model and establish a single, universal physical explanation for UMR. By doing so, the work offers a simpler and more comprehensive way to understand magnetoresistance across a wide range of spintronic systems.

This research was recently published in National Science Review under the title "Physics Origin of Universal Unusual Magnetoresistance."

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Journal Reference:

1. Lijun Zhu, Qianbiao Liu, Xiangrong Wang. **Physics origin of universal unusual magnetoresistance**. *National Science Review*, 2025; 12 (8) DOI: [10.1093/nsr/nwaf240](https://doi.org/10.1093/nsr/nwaf240)
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